where δ_{ij} is the usual Kronecker symbol.

Next, let's take |Φ(0)> and |Ψ(0)> to be members of this orthonormal basis. Substituting into Eq. 4.4 gives

$$\langle i|\mathbf{U}^{\dagger}(t)\mathbf{U}(t)|j\rangle = 0 \quad (i \neq j)$$

whenever *i* and *j* are not the same. On the other hand, if *i* and *j* are the same, then so are the output vectors **U**(*t*)|*i*> and **U**(*t*)|*j*>. In that case, the inner product between them should be 1. Therefore, the general relation takes the form

$$\langle i|\mathbf{U}^{\dagger}(t)\mathbf{U}(t)|j\rangle = \delta_{ij}.$$

In other words, the operator **U**†(*t*)**U**(*t*) behaves like the unit operator *I* when it acts between any members of a basis set. From here it is an easy matter to prove that **U**†(*t*)**U**(*t*) acts like the unit operator *I* when it acts on any state. An operator **U** that satisfies

$$\mathbf{U}^{\dagger}\mathbf{U} = I$$

is called *unitary*. In physics lingo, *time evolution is unitary*.

Unitary operators play an enormous role in quantum mechanics, representing all sorts of transformations on the state-space. Time evolution is just one example. Thus, we conclude this section with a fifth principle of quantum mechanics:

- Principle 5: The evolution of state-vectors with time is unitary.

Exercise 4.1: Prove that if **U** is unitary, and if |*A*> and |*B*> are any two state-vectors, then the inner product of **U**|*A*> and **U**|*B*> is the same as the inner product of |*A*> and |*B*>. One could call this the *conservation*